3.2.3 Brain and neuropsychology

Content	Additional information
Structure and function of the nervous system	The divisions of the human nervous system: central and peripheral (somatic and autonomic), basic functions of these divisions. The autonomic nervous system and the fight or flight response. The James-Lange theory of emotion.
Neuron structure and function	Sensory, relay and motor neurons. Synaptic transmission: release and reuptake of neurotransmitters. Excitation and inhibition. An understanding of how these processes interact. Hebb's theory of learning and neuronal growth.
Structure and function of the brain	Brain structure: frontal lobe, temporal lobe, parietal lobe, occipital lobe and cerebellum. Basic function of these structures. Localisation of function in the brain: motor, somatosensory, visual, auditory and language areas. Penfield’s study of the interpretive cortex.

Content	Additional information
An introduction to neuropsychology	Cognitive neuroscience: how the structure and function of the brain relate to behaviour and cognition. The use of scanning techniques to identify brain functioning: CT, PET and fMRI scans. Tulving's 'gold' memory study. A basic understanding of how neurological damage, eg stroke or injury can affect motor abilities and behaviour.